

Chapter 2

Math Refresher

This chapter reviews the concepts from probability and optimization that will be useful at various points in the course. The emphasis is on finite and countable probability spaces, with a brief introduction to continuous random variables and Gaussian distributions. We avoid measure-theoretic details and focus on the identities and calculations that recur in algorithmic game theory.

2.1 Probability Refresher

Probability provides a mathematical language for describing and reasoning about uncertainty. Uncertainty may arise from genuinely random phenomena, or it may be useful to model a complicated process as random when tracking all of its underlying details is impractical. For example, the outcome of a coin toss is determined by its physical conditions, but modeling it as a random experiment gives a simple and useful description.

Probability allows us to quantify uncertainty, compare likely outcomes, and reason about averages and rare events. These ideas will be used repeatedly throughout the course when studying randomized strategies, uncertain information, and learning from observations.

2.1.1 Probability spaces, outcomes, and events

A probability model consists of a sample space Ω of possible outcomes together with probabilities assigned to events.

Definition 2.1 (Outcome and event). An *outcome* is a single element $\omega \in \Omega$. An *event* is a subset $A \subseteq \Omega$.

For example, if two fair coins are tossed, then

$$\Omega = \{HH, HT, TH, TT\}.$$

The event “exactly one head” is

$$A = \{HT, TH\}.$$

For a finite or countable sample space, probabilities satisfy

$$\mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\omega), \quad \sum_{\omega \in \Omega} \mathbb{P}(\omega) = 1.$$

2.1.2 Random variables

A random variable is a function that assigns a number to each outcome.

Definition 2.2 (Random variable). A random variable is a function

$$X : \Omega \rightarrow \mathbb{R}.$$

For example, for two coin tosses define X to be the number of heads. Then

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

Discrete versus continuous random variables

The distinction between discrete and continuous refers to the *distribution of X* , not to whether the underlying sample space Ω is discrete or continuous.

Example 2.3 (Continuous sample space, discrete random variable). Let $\Omega = [0, 1]$ with the uniform probability distribution, and define

$$X(\omega) = \begin{cases} -1, & \omega \leq \frac{1}{2}, \\ 1, & \omega > \frac{1}{2}. \end{cases}$$

Although Ω is continuous, X takes only two values. Hence X is discrete, with

$$\mathbb{P}(X = -1) = \mathbb{P}(X = 1) = \frac{1}{2}.$$

Example 2.4 (Continuous random variable on the same space). Again let $\Omega = [0, 1]$ be uniform, but now define

$$X(\omega) = \omega.$$

Then $X \sim \text{Uniform}[0, 1]$, so X is a continuous random variable.

For a discrete random variable, we write its probability mass function as

$$p_X(x) = \mathbb{P}(X = x).$$

The probabilities satisfy

$$\sum_x p_X(x) = 1.$$

2.1.3 Joint distributions and marginalization

Suppose X and Y are random variables defined on the same probability space. Their joint distribution is

$$p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y).$$

If we know the joint distribution but only care about X , we sum out the possible values of Y :

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

and similarly

$$p_Y(y) = \sum_x p_{X,Y}(x, y).$$

This operation is called *marginalization*.

Example 2.5. Suppose

	$Y = 0$	$Y = 1$
$X = 0$	0.2	0.3
$X = 1$	0.1	0.4

Then

$$\mathbb{P}(X = 0) = 0.2 + 0.3 = 0.5, \quad \mathbb{P}(X = 1) = 0.1 + 0.4 = 0.5.$$

2.2 Expectation

For a discrete random variable X , its expectation is

$$\mathbb{E}[X] = \sum_x x \mathbb{P}(X = x)$$

whenever the sum is well defined. For a countably supported random variable, a sufficient condition is $\mathbb{E}[|X|] < \infty$.

The lazy statistician rule (LOTUS)

Suppose we want the expectation of a function $g(X)$. We do not need to first derive the distribution of $g(X)$. Instead,

$$\mathbb{E}[g(X)] = \sum_x g(x) \mathbb{P}(X = x)$$

for a discrete random variable. This identity is commonly called the *Law of the Unconscious Statistician* (LOTUS); informally, it is the “lazy statistician rule.”

To see why it holds, start from the definition of the expectation of $g(X)$:

$$\mathbb{E}[g(X)] = \sum_z z \mathbb{P}(g(X) = z).$$

Now group together all values x for which $g(x) = z$:

$$\mathbb{P}(g(X) = z) = \sum_{x: g(x)=z} \mathbb{P}(X = x).$$

Hence

$$\begin{aligned} \mathbb{E}[g(X)] &= \sum_z z \sum_{x: g(x)=z} \mathbb{P}(X = x) \\ &= \sum_x g(x) \mathbb{P}(X = x). \end{aligned}$$

Example 2.6. If X takes values $-1, 0, 1$ uniformly, then

$$\mathbb{E}[X^2] = (-1)^2 \frac{1}{3} + 0^2 \frac{1}{3} + 1^2 \frac{1}{3} = \frac{2}{3}.$$

We did not need to separately compute the distribution of X^2 .

Indicators and counting An indicator random variable for an event A is

$$\mathbf{1}_A = \begin{cases} 1, & A \text{ occurs,} \\ 0, & A \text{ does not occur.} \end{cases}$$

Since $\mathbf{1}_A$ is Bernoulli,

$$\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A).$$

This makes indicator variables extremely useful for counting.

2.2.1 Variance and covariance

Let $\mu = \mathbb{E}[X]$. The variance of X is

$$\boxed{\text{Var}(X) = \mathbb{E}[(X - \mu)^2].}$$

Expanding the square gives the useful identity

$$\begin{aligned} \text{Var}(X) &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \boxed{\mathbb{E}[X^2] - \mathbb{E}[X]^2}. \end{aligned}$$

For two random variables X, Y , their covariance is

$$\boxed{\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])].}$$

Expanding,

$$\begin{aligned} \text{Cov}(X, Y) &= \mathbb{E}[XY - X\mathbb{E}[Y] - Y\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y]] \\ &= \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] - \mathbb{E}[Y]\mathbb{E}[X] + \mathbb{E}[X]\mathbb{E}[Y] \\ &= \boxed{\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]}. \end{aligned}$$

Two random variables are *uncorrelated* if

$$\text{Cov}(X, Y) = 0,$$

or equivalently

$$\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y].$$

This does not generally mean $\mathbb{E}[XY] = 0$ unless at least one of the means is zero.

Independence implies zero covariance

If X and Y are independent, then their joint mass function factorizes:

$$p_{X,Y}(x, y) = p_X(x)p_Y(y).$$

Therefore

$$\begin{aligned} \mathbb{E}[XY] &= \sum_x \sum_y xy p_{X,Y}(x, y) \\ &= \sum_x \sum_y xy p_X(x)p_Y(y) \\ &= \left(\sum_x xp_X(x) \right) \left(\sum_y yp_Y(y) \right) \\ &= \mathbb{E}[X]\mathbb{E}[Y]. \end{aligned}$$

Hence

$$\boxed{X \perp Y \implies \text{Cov}(X, Y) = 0.}$$

The converse is false in general.

Gaussian exception. For jointly Gaussian random variables, zero covariance *does* imply independence. Thus if (X, Y) is jointly Gaussian,

$$\text{Cov}(X, Y) = 0 \iff X \perp Y.$$

The assumption that they are *jointly* Gaussian is important; it is not enough for the two marginal distributions to be Gaussian separately.

2.2.2 Conditional probability

For events A and B with $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditioning means that we restrict our attention to outcomes in which B has occurred and renormalize probabilities within that restricted set.

For random variables,

$$\mathbb{P}(X = x | Y = y) = \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(Y = y)}, \quad \mathbb{P}(Y = y) > 0.$$

Law of total probability

If $B_1, \dots, B_m$ partition the sample space and $\mathbb{P}(B_i) > 0$, then

$$\mathbb{P}(A) = \sum_{i=1}^m \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

Indeed,

$$\begin{aligned} \mathbb{P}(A) &= \sum_{i=1}^m \mathbb{P}(A \cap B_i) \\ &= \sum_{i=1}^m \mathbb{P}(A | B_i) \mathbb{P}(B_i). \end{aligned}$$

Bayes' rule

For events A, B with $\mathbb{P}(A) > 0$ and $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A) \mathbb{P}(A)}{\mathbb{P}(B)}.$$

This follows because

$$\mathbb{P}(A \cap B) = \mathbb{P}(A | B) \mathbb{P}(B) = \mathbb{P}(B | A) \mathbb{P}(A).$$

A useful interpretation is:

$$\text{posterior} = \frac{\text{likelihood} \times \text{prior}}{\text{evidence}}.$$

Here

- $\mathbb{P}(A)$ is the *prior*: our belief in hypothesis A before seeing B ;
- $\mathbb{P}(B | A)$ is the *likelihood*: how plausible the observed evidence B would be if A were true;

- $\mathbb{P}(B)$ is the *evidence* or *marginal likelihood*: how likely B is overall, across all hypotheses;
- $\mathbb{P}(A \mid B)$ is the *posterior*: our updated belief in A after observing B .

If A and A^c are the two possibilities, then

$$\mathbb{P}(B) = \mathbb{P}(B \mid A)\mathbb{P}(A) + \mathbb{P}(B \mid A^c)\mathbb{P}(A^c).$$

Thus the denominator normalizes the updated belief by accounting for all ways in which the evidence could have arisen.

2.2.3 Conditional expectation

For discrete random variables X, Y and a value y with $\mathbb{P}(Y = y) > 0$,

$$\mathbb{E}[X \mid Y = y] = \sum_x x \mathbb{P}(X = x \mid Y = y).$$

This is simply the expectation of X computed under the conditional distribution given $Y = y$.

The quantity $\mathbb{E}[X \mid Y]$ is itself a random variable: once the value of Y is known, it returns the corresponding conditional expectation.

Law of total expectation

For finite discrete random variables,

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]] = \sum_y \mathbb{E}[X \mid Y = y] \mathbb{P}(Y = y).$$

Proof. Starting from the definition,

$$\mathbb{E}[X] = \sum_x x \mathbb{P}(X = x).$$

Marginalize over Y :

$$\mathbb{P}(X = x) = \sum_y \mathbb{P}(X = x, Y = y).$$

Therefore

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x \sum_y \mathbb{P}(X = x, Y = y) \\ &= \sum_y \sum_x x \mathbb{P}(X = x, Y = y) \\ &= \sum_y \sum_x x \mathbb{P}(X = x \mid Y = y) \mathbb{P}(Y = y) \\ &= \sum_y \mathbb{P}(Y = y) \underbrace{\sum_x x \mathbb{P}(X = x \mid Y = y)}_{\mathbb{E}[X \mid Y=y]} \\ &= \sum_y \mathbb{E}[X \mid Y = y] \mathbb{P}(Y = y) \\ &= \mathbb{E}[\mathbb{E}[X \mid Y]]. \end{aligned}$$

For countable supports, the same proof works whenever the relevant sums are absolutely convergent, for example when $\mathbb{E}[|X|] < \infty$.

2.2.4 Continuous random variables

A continuous random variable is described by a probability density function f_X satisfying

$$f_X(x) \geq 0, \quad \int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Probabilities are obtained by integration:

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f_X(x) dx.$$

For a continuous random variable,

$$\mathbb{P}(X = x) = 0$$

for every individual point x .

The expectation is

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx.$$

The continuous version of LOTUS is

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx.$$

For jointly continuous X, Y with joint density $f_{X,Y}$, marginalization becomes

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

2.2.5 Gaussian random variables

Univariate Gaussian

A Gaussian random variable with mean μ and variance σ^2 is written

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

and has density

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

The mean μ controls the center, while σ controls the spread.

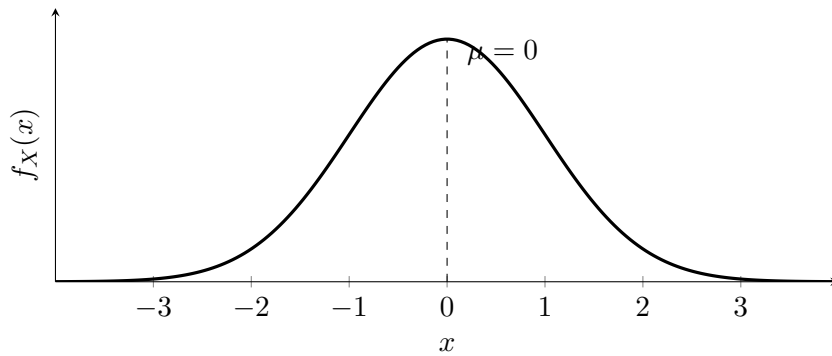


Figure 2.1: The density of a standard Gaussian $\mathcal{N}(0, 1)$.

Multivariate Gaussian

A random vector $X \in \mathbb{R}^d$ is multivariate Gaussian with mean vector $\mu \in \mathbb{R}^d$ and positive-definite covariance matrix $\Sigma \in \mathbb{R}^{d \times d}$ if

$$X \sim \mathcal{N}(\mu, \Sigma)$$

with density

$$f_X(x) = \frac{1}{(2\pi)^{d/2} |\Sigma|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu)^\top \Sigma^{-1} (x - \mu) \right).$$

The covariance matrix controls both the spread in each coordinate and the dependence between coordinates.

In two dimensions,

$$\Sigma = \begin{pmatrix} \sigma_1^2 & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_1, X_2) & \sigma_2^2 \end{pmatrix}.$$

If the off-diagonal covariance is zero, the contours are axis-aligned. A nonzero covariance rotates the elliptical contours.

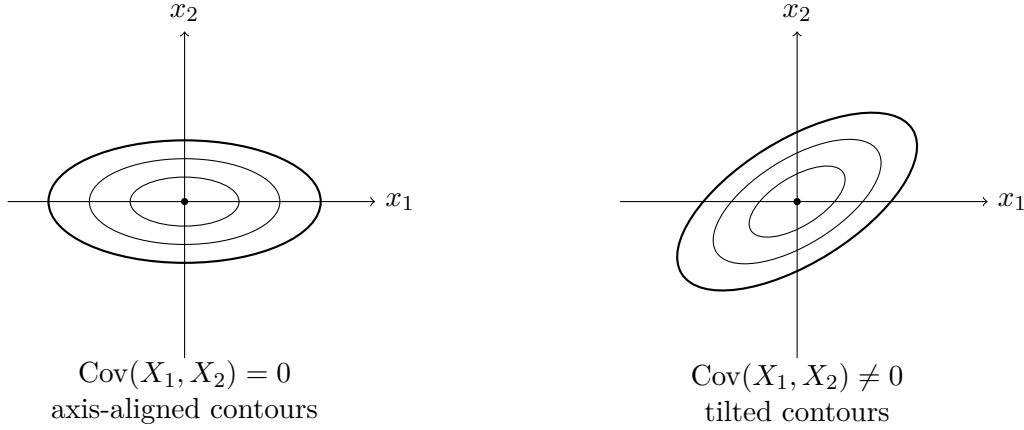


Figure 2.2: Schematic level sets of two-dimensional Gaussian densities. For jointly Gaussian coordinates, zero covariance implies independence.

2.2.6 Markov and Chebyshev inequalities

These inequalities convert information about expectations or variances into probability bounds.

Markov's inequality

If $X \geq 0$ and $a > 0$, then

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

For a nonnegative discrete random variable,

$$\begin{aligned} \mathbb{E}[X] &= \sum_x x \mathbb{P}(X = x) \\ &\geq \sum_{x \geq a} x \mathbb{P}(X = x) \\ &\geq \sum_{x \geq a} a \mathbb{P}(X = x) \\ &= a \mathbb{P}(X \geq a). \end{aligned}$$

Dividing by a gives the result.

Chebyshev's inequality

Let $\mu = \mathbb{E}[X]$ and $\sigma^2 = \text{Var}(X)$. For $k > 0$,

$$\mathbb{P}(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}.$$

The key observation is

$$|X - \mu| \geq k\sigma \iff (X - \mu)^2 \geq k^2\sigma^2.$$

Apply Markov's inequality to the nonnegative random variable

$$Y = (X - \mu)^2.$$

Then

$$\begin{aligned} \mathbb{P}(|X - \mu| \geq k\sigma) &= \mathbb{P}((X - \mu)^2 \geq k^2\sigma^2) \\ &\leq \frac{\mathbb{E}[(X - \mu)^2]}{k^2\sigma^2} \\ &= \frac{\sigma^2}{k^2\sigma^2} \\ &= \frac{1}{k^2}. \end{aligned}$$

Thus Chebyshev's inequality is simply Markov's inequality applied to the squared deviation from the mean.

2.2.7 Questions

Exercise 2.7. For any random variables X, Y with finite expectations and constants a, b , show that:

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].$$

Importantly, X and Y do *not* need to be independent.

Exercise 2.8. Suppose there are n students. Each student independently answers a given question correctly with probability p .

- (a) What is the expected number of students who answer the question correctly?
- (b) A teaching assistant checks the answers one by one, stopping as soon as a correct answer is found, or after all n answers have been checked. What is the expected number of answers that the teaching assistant checks?
- (c) Now suppose the instructor randomly chooses between two possible questions:
 - an *easy* question, chosen with probability q , for which each student answers correctly with probability p_E ;
 - a *hard* question, chosen with probability $1 - q$, for which each student answers correctly with probability p_H .

Let X denote the total number of students who answer correctly. Compute $\mathbb{E}[X]$ using conditional expectations and the law of total expectation.

Hint. Define appropriate indicator random variables for each part.

Exercise 2.9. Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Verify directly from the joint distribution that X and Y are not independent, but are uncorrelated.

Exercise 2.10. A disease affects 1% of a population. A test is positive with probability 0.99 when the disease is present and with probability 0.05 when it is absent. Compute the probability that a person has the disease given a positive test.

Exercise 2.11. Let $X \sim \mathcal{N}(0, 1)$ and define $Y = 2X + 3$. What are $\mathbb{E}[Y]$ and $\text{Var}(Y)$? What is the distribution of Y ?

2.3 Optimization Refresher

Optimization asks us to choose a feasible decision that minimizes (or maximizes) an objective. Many problems in algorithmic game theory can be expressed in this language: a mixed strategy is a point in a simplex, best-response problems are optimization problems, and several equilibrium and learning problems can be studied using convex optimization. Convexity is especially useful because it provides strong structural guarantees and admits efficient algorithms in many important settings. A standard reference is [BV04]; see also [NW06] for numerical optimization.

Throughout this section, unless stated otherwise, we work in $\mathbb{R}^d$ and use the Euclidean inner product $\langle \cdot, \cdot \rangle$ and norm $\|\cdot\|_2$. Bold symbols are used to indicate vectors.

2.3.1 Convex sets

A convex set contains the entire line segment joining any two of its points.

Definition 2.12 (Convex set). A set $\mathcal{X} \subseteq \mathbb{R}^d$ is *convex* if, for every $\mathbf{x}, \mathbf{y} \in \mathcal{X}$ and every $\lambda \in [0, 1]$,

$$\lambda \mathbf{x} + (1 - \lambda) \mathbf{y} \in \mathcal{X}.$$

The point $\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}$ is a *convex combination* of $\mathbf{x}$ and $\mathbf{y}$. More generally, if $\lambda_i \geq 0$ and $\sum_{i=1}^m \lambda_i = 1$, then

$$\sum_{i=1}^m \lambda_i \mathbf{x}_i$$

is a convex combination of the points $\mathbf{x}_1, \dots, \mathbf{x}_m$.

Important examples of convex sets include affine subspaces, half-spaces, Euclidean balls, boxes, polytopes, and the probability simplex

$$\Delta_d := \left\{ \mathbf{x} \in \mathbb{R}^d : x_i \geq 0 \ \forall i, \sum_{i=1}^d x_i = 1 \right\}.$$

The simplex will appear repeatedly when we talk about strategies for games later in the course.

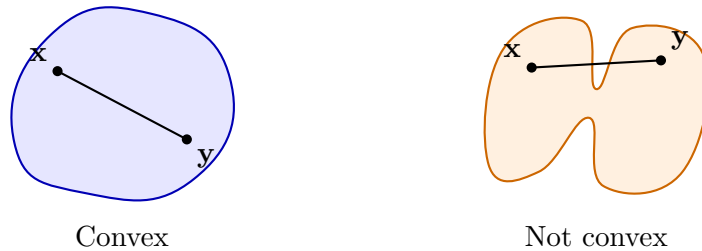


Figure 2.3: In a convex set, every line segment between two feasible points remains feasible.

2.3.2 Convex functions

To define convexity of a function, we assume that its domain is convex.

Definition 2.13 (Convex function). Let $\mathcal{X} \subseteq \mathbb{R}^d$ be convex. A function $h : \mathcal{X} \rightarrow \mathbb{R}$ is *convex* if, for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$ and $\lambda \in [0, 1]$,

$$h(\lambda \mathbf{x} + (1 - \lambda) \mathbf{y}) \leq \lambda h(\mathbf{x}) + (1 - \lambda) h(\mathbf{y}).$$

Geometrically, the value of a convex function on a line segment lies below the straight chord joining the function values at the endpoints.

A function h is *concave* if $-h$ is convex. It is *strictly convex* if the defining inequality is strict whenever $\mathbf{x} \neq \mathbf{y}$ and $\lambda \in (0, 1)$.

Example 2.14. The functions

$$h(x) = x^2, \quad h(\mathbf{x}) = \|\mathbf{x}\|_2^2, \quad h(x) = e^x$$

are convex. A linear function $h(\mathbf{x}) = \langle \mathbf{c}, \mathbf{x} \rangle$ is both convex and concave.

Jensen's inequality

The two-point definition immediately extends to any finite convex combination.

Theorem 2.15 (Jensen's inequality: finite form). Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be convex. Let $\mathbf{x}_1, \dots, \mathbf{x}_m \in \mathcal{X}$ and let $\lambda_i \geq 0$ satisfy $\sum_{i=1}^m \lambda_i = 1$. Then

$$h\left(\sum_{i=1}^m \lambda_i \mathbf{x}_i\right) \leq \sum_{i=1}^m \lambda_i h(\mathbf{x}_i).$$

Proof. We prove the claim by induction on m . The case $m = 2$ is exactly the definition of convexity. Assume the claim holds for $m - 1$ points. Let

$$\alpha := \sum_{i=1}^{m-1} \lambda_i = 1 - \lambda_m.$$

If $\alpha = 0$, then $\lambda_m = 1$ and the claim is trivial. Otherwise define

$$\mathbf{z} := \sum_{i=1}^{m-1} \frac{\lambda_i}{\alpha} \mathbf{x}_i.$$

The coefficients in this sum are nonnegative and sum to one, so by the induction hypothesis,

$$h(\mathbf{z}) \leq \sum_{i=1}^{m-1} \frac{\lambda_i}{\alpha} h(\mathbf{x}_i).$$

Since

$$\sum_{i=1}^m \lambda_i \mathbf{x}_i = \alpha \mathbf{z} + \lambda_m \mathbf{x}_m,$$

two-point convexity gives

$$\begin{aligned} h\left(\sum_{i=1}^m \lambda_i \mathbf{x}_i\right) &\leq \alpha h(\mathbf{z}) + \lambda_m h(\mathbf{x}_m) \\ &\leq \sum_{i=1}^{m-1} \lambda_i h(\mathbf{x}_i) + \lambda_m h(\mathbf{x}_m), \end{aligned}$$

which proves the result. □

This has an immediate probabilistic interpretation.

Corollary 2.16 (Jensen's inequality for a discrete random variable). *If X is a finitely supported random variable and h is convex, then*

$$\boxed{h(\mathbb{E}[X]) \leq \mathbb{E}[h(X)]}.$$

Proof. Suppose X takes values $x_1, \dots, x_m$ and let $p_i = \mathbb{P}(X = x_i)$. Then $p_i \geq 0$ and $\sum_i p_i = 1$. Hence

$$h(\mathbb{E}[X]) = h\left(\sum_{i=1}^m p_i x_i\right) \leq \sum_{i=1}^m p_i h(x_i) = \mathbb{E}[h(X)].$$

□

The same statement holds much more generally whenever the expectations are well defined. This version will occasionally be useful when randomization and convex objectives interact.

2.3.3 First-order characterizations of convexity

For differentiable functions, convexity can be characterized by tangent hyperplanes.

Theorem 2.17 (First-order characterization). *Let $\mathcal{X} \subseteq \mathbb{R}^d$ be convex and let $h : \mathcal{X} \rightarrow \mathbb{R}$ be differentiable. Then h is convex if and only if, for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$,*

$$\boxed{h(\mathbf{x}) \geq h(\mathbf{y}) + \langle \nabla h(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle}.$$

Thus, for a differentiable convex function, the tangent hyperplane at any point is a global lower bound on the function.

Proof. First suppose that h is convex. Fix $\mathbf{x}, \mathbf{y} \in \mathcal{X}$ and define

$$\phi(t) := h(\mathbf{y} + t(\mathbf{x} - \mathbf{y})), \quad t \in [0, 1].$$

By convexity,

$$\phi(t) \leq (1 - t)\phi(0) + t\phi(1).$$

Rearranging,

$$\phi(1) - \phi(0) \geq \frac{\phi(t) - \phi(0)}{t}.$$

Letting $t \downarrow 0$ gives

$$h(\mathbf{x}) - h(\mathbf{y}) \geq \phi'(0) = \langle \nabla h(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle.$$

Conversely, suppose the displayed first-order inequality holds. Let

$$\mathbf{z} = \lambda \mathbf{x} + (1 - \lambda)\mathbf{y}, \quad \lambda \in [0, 1].$$

Apply the first-order inequality once with $(\mathbf{x}, \mathbf{z})$ and once with $(\mathbf{y}, \mathbf{z})$:

$$\begin{aligned} h(\mathbf{x}) &\geq h(\mathbf{z}) + \langle \nabla h(\mathbf{z}), \mathbf{x} - \mathbf{z} \rangle, \\ h(\mathbf{y}) &\geq h(\mathbf{z}) + \langle \nabla h(\mathbf{z}), \mathbf{y} - \mathbf{z} \rangle. \end{aligned}$$

Multiply the first inequality by λ , the second by $1 - \lambda$, and add them. Because

$$\lambda(\mathbf{x} - \mathbf{z}) + (1 - \lambda)(\mathbf{y} - \mathbf{z}) = \mathbf{0},$$

the gradient terms cancel, giving

$$\lambda h(\mathbf{x}) + (1 - \lambda)h(\mathbf{y}) \geq h(\mathbf{z}).$$

This is exactly convexity. □

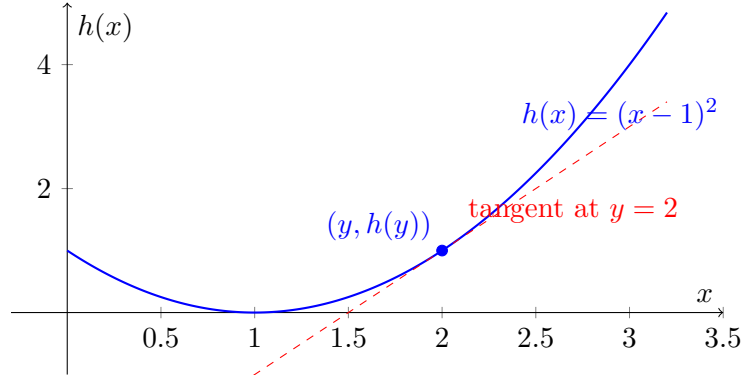


Figure 2.4: For a differentiable convex function, every tangent line lies below the graph.

2.3.4 Strong convexity and smoothness

Convexity can be strengthened by requiring a uniform amount of curvature.

Definition 2.18 (Strong convexity). A differentiable function $h : \mathcal{X} \rightarrow \mathbb{R}$ is α -strongly convex, for $\alpha > 0$, if for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$,

$$h(\mathbf{x}) \geq h(\mathbf{y}) + \langle \nabla h(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle + \frac{\alpha}{2} \|\mathbf{x} - \mathbf{y}\|_2^2.$$

Strong convexity says that the function lies not merely above its tangent hyperplane, but above a quadratic bowl around that tangent hyperplane.

A different property controls how quickly the gradient can change.

Definition 2.19 (Smoothness). A differentiable function $h : \mathcal{X} \rightarrow \mathbb{R}$ is L -smooth if

$$\|\nabla h(\mathbf{x}) - \nabla h(\mathbf{y})\|_2 \leq L \|\mathbf{x} - \mathbf{y}\|_2$$

for all $\mathbf{x}, \mathbf{y} \in \mathcal{X}$.

For an L -smooth function one has the useful upper quadratic bound

$$h(\mathbf{x}) \leq h(\mathbf{y}) + \langle \nabla h(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle + \frac{L}{2} \|\mathbf{x} - \mathbf{y}\|_2^2.$$

Thus, strong convexity gives a quadratic *lower* bound and smoothness gives a quadratic *upper* bound.

2.3.5 Second-order conditions

When a function is twice differentiable, its Hessian provides a convenient way to test convexity.

Theorem 2.20 (Second-order characterization). Suppose h is twice differentiable on an open convex set containing $\mathcal{X}$. Then

$$h \text{ is convex} \iff \nabla^2 h(\mathbf{x}) \succeq 0 \quad \text{for every } \mathbf{x} \in \mathcal{X}.$$

Here $\nabla^2 h(\mathbf{x}) \succeq 0$ means

$$\mathbf{v}^\top \nabla^2 h(\mathbf{x}) \mathbf{v} \geq 0 \quad \text{for every } \mathbf{v} \in \mathbb{R}^d.$$

Proof. Fix $\mathbf{x}, \mathbf{y} \in \mathcal{X}$ and define the one-dimensional restriction

$$\phi(t) := h(\mathbf{x} + t(\mathbf{y} - \mathbf{x})), \quad t \in [0, 1].$$

By the chain rule,

$$\phi''(t) = (\mathbf{y} - \mathbf{x})^\top \nabla^2 h(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) (\mathbf{y} - \mathbf{x}).$$

If the Hessian is positive semidefinite everywhere, then $\phi''(t) \geq 0$, so ϕ is a convex one-dimensional function on every line segment. Hence h is convex.

Conversely, if h is convex, then every such one-dimensional restriction ϕ is convex. For a twice differentiable one-dimensional convex function, $\phi''(t) \geq 0$. Taking an arbitrary point $\mathbf{x}$ and direction $\mathbf{v}$ gives

$$\mathbf{v}^\top \nabla^2 h(\mathbf{x}) \mathbf{v} \geq 0,$$

so the Hessian is positive semidefinite. □

The same idea gives the following useful tests:

$$\nabla^2 h(\mathbf{x}) \succeq \alpha I \implies h \text{ is } \alpha\text{-strongly convex,}$$

and, for twice differentiable convex h ,

$$0 \preceq \nabla^2 h(\mathbf{x}) \preceq LI \iff h \text{ is } L\text{-smooth.}$$

Example 2.21. For

$$h(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top Q \mathbf{x} + \mathbf{c}^\top \mathbf{x} + r,$$

where $Q = Q^\top$, we have

$$\nabla h(\mathbf{x}) = Q\mathbf{x} + \mathbf{c}, \quad \nabla^2 h(\mathbf{x}) = Q.$$

Hence h is convex exactly when $Q \succeq 0$, and it is α -strongly convex when $Q \succeq \alpha I$.

2.3.6 Why convexity is useful: local and global minima

The central structural fact is that convex functions do not have “bad” local minima.

Theorem 2.22 (Local minima are global). *Let $\mathcal{X}$ be convex and let $h : \mathcal{X} \rightarrow \mathbb{R}$ be convex. Every local minimizer of h over $\mathcal{X}$ is a global minimizer.*

Proof. Let $\mathbf{x}^*$ be a local minimizer. Suppose, for contradiction, that it is not global. Then there exists $\mathbf{y} \in \mathcal{X}$ such that

$$h(\mathbf{y}) < h(\mathbf{x}^*).$$

For $t \in (0, 1)$ define

$$\mathbf{x}_t = (1 - t)\mathbf{x}^* + t\mathbf{y}.$$

Because $\mathcal{X}$ is convex, $\mathbf{x}_t \in \mathcal{X}$, and for sufficiently small $t > 0$ the point $\mathbf{x}_t$ lies in any prescribed neighborhood of $\mathbf{x}^*$. By convexity,

$$h(\mathbf{x}_t) \leq (1 - t)h(\mathbf{x}^*) + th(\mathbf{y}) < h(\mathbf{x}^*).$$

This contradicts local optimality. Therefore $\mathbf{x}^*$ is global. □

Notice that compactness is *not* needed for Theorem 2.22. Compactness is relevant to the separate question of whether a minimizer exists.

Theorem 2.23 (Existence on a compact domain). *If $\mathcal{X}$ is nonempty and compact and $h : \mathcal{X} \rightarrow \mathbb{R}$ is continuous, then h attains a minimum and a maximum on $\mathcal{X}$.*

In our applications, functions are continuous and usually differentiable, so compact feasible regions such as simplices and bounded polytopes guarantee existence.

Strict convexity gives uniqueness.

Theorem 2.24 (Uniqueness under strict convexity). *If h is strictly convex on a convex set $\mathcal{X}$, then h has at most one global minimizer. Consequently, if a minimizer exists, it is unique.*

Proof. Suppose there are two distinct minimizers $\mathbf{x}_1 \neq \mathbf{x}_2$ with common optimal value h^* . By strict convexity,

$$h\left(\frac{\mathbf{x}_1 + \mathbf{x}_2}{2}\right) < \frac{1}{2}h(\mathbf{x}_1) + \frac{1}{2}h(\mathbf{x}_2) = h^*,$$

contradicting the definition of h^* . $\square$

Every strongly convex function is strictly convex, and therefore has at most one minimizer.

2.3.7 First-order optimality over a convex set

For constrained convex optimization, the gradient need not be zero at an optimum: the boundary may prevent us from moving in the direction of $-\nabla h$. The correct condition is a variational inequality.

Theorem 2.25 (First-order optimality condition). *Let $\mathcal{X}$ be convex and let $h : \mathcal{X} \rightarrow \mathbb{R}$ be differentiable and convex. A point $\mathbf{x}^* \in \mathcal{X}$ is a global minimizer if and only if*

$$\boxed{\langle \nabla h(\mathbf{x}^*), \mathbf{x} - \mathbf{x}^* \rangle \geq 0 \quad \text{for every } \mathbf{x} \in \mathcal{X}.}$$

Proof. First suppose the displayed inequality holds. By the first-order characterization of convexity,

$$h(\mathbf{x}) \geq h(\mathbf{x}^*) + \langle \nabla h(\mathbf{x}^*), \mathbf{x} - \mathbf{x}^* \rangle \geq h(\mathbf{x}^*)$$

for every $\mathbf{x} \in \mathcal{X}$. Hence $\mathbf{x}^*$ is globally optimal.

Conversely, suppose $\mathbf{x}^*$ is globally optimal. Fix any $\mathbf{x} \in \mathcal{X}$ and define

$$\phi(t) = h(\mathbf{x}^* + t(\mathbf{x} - \mathbf{x}^*)), \quad t \in [0, 1].$$

The point $t = 0$ minimizes ϕ over $[0, 1]$, so its right derivative satisfies $\phi'(0) \geq 0$. By the chain rule,

$$\phi'(0) = \langle \nabla h(\mathbf{x}^*), \mathbf{x} - \mathbf{x}^* \rangle,$$

which proves the condition. $\square$

If $\mathbf{x}^*$ is an interior point of $\mathcal{X}$, then we may move a small amount in both directions along every coordinate. The condition above then reduces to

$$\nabla h(\mathbf{x}^*) = \mathbf{0}.$$

2.3.8 Gradient descent and projected gradient descent

For an unconstrained differentiable objective, gradient descent repeatedly moves opposite to the gradient:

$$\boxed{\mathbf{x}^{t+1} = \mathbf{x}^t - \eta_t \nabla h(\mathbf{x}^t).}$$

The step size $\eta_t > 0$ controls how far we move.

When the feasible set is a proper subset $\mathcal{X} \subsetneq \mathbb{R}^d$, a gradient step may leave the feasible region. Projected gradient descent corrects this by projecting back:

$$\boxed{\mathbf{x}^{t+1} = \Pi_{\mathcal{X}}(\mathbf{x}^t - \eta_t \nabla h(\mathbf{x}^t)),}$$

where

$$\Pi_{\mathcal{X}}(\mathbf{z}) := \arg \min_{\mathbf{x} \in \mathcal{X}} \|\mathbf{x} - \mathbf{z}\|_2.$$

For closed convex $\mathcal{X}$, the Euclidean projection exists and is unique.

For smooth convex objectives, appropriate step sizes yield convergence to a global minimizer. The detailed convergence-rate analysis is not needed in this refresher, but the update will reappear later in online learning.

2.3.9 Constrained optimization: the Lagrangian and KKT conditions

Consider the constrained minimization problem

$$\begin{aligned} \min_{\mathbf{x} \in \mathbb{R}^d} \quad & h(\mathbf{x}) \\ \text{subject to} \quad & g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \\ & a_j(\mathbf{x}) = 0, \quad j = 1, \dots, r. \end{aligned} \tag{2.1}$$

We associate a nonnegative multiplier λ_i with each inequality constraint and an unrestricted multiplier ν_j with each equality constraint.

Definition 2.26 (Lagrangian). The *Lagrangian* of (2.1) is

$$\mathcal{L}(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = h(\mathbf{x}) + \sum_{i=1}^m \lambda_i g_i(\mathbf{x}) + \sum_{j=1}^r \nu_j a_j(\mathbf{x}).$$

At an optimum, the Lagrange multipliers measure how the active constraints balance the gradient of the objective.

Theorem 2.27 (Karush–Kuhn–Tucker conditions). *Suppose h and $g_1, \dots, g_m$ are differentiable convex functions and the equality constraints a_j are affine. Under a standard constraint qualification such as Slater’s condition, a feasible point $\mathbf{x}^*$ is optimal if and only if there exist multipliers $\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*$ satisfying:*

(i) **Primal feasibility:**

$$g_i(\mathbf{x}^*) \leq 0, \quad a_j(\mathbf{x}^*) = 0.$$

(ii) **Dual feasibility:**

$$\lambda_i^* \geq 0.$$

(iii) **Stationarity:**

$$\nabla h(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^r \nu_j^* \nabla a_j(\mathbf{x}^*) = \mathbf{0}.$$

(iv) **Complementary slackness:**

$$\lambda_i^* g_i(\mathbf{x}^*) = 0 \quad \forall i.$$

Slater’s condition means, roughly, that there exists a feasible point satisfying all nonlinear inequality constraints *strictly*. It is a convenient condition that rules out degenerate pathologies. The full necessity proof of the KKT theorem uses convex duality/separation results and is beyond this refresher. The sufficiency direction, however, is simple and useful.

Proof that the KKT conditions are sufficient in the convex case. Fix multipliers λ^*, ν^* satisfying the KKT conditions. Because h and the g_i are convex, the function

$$\mathbf{x} \mapsto \mathcal{L}(\mathbf{x}, \lambda^*, \nu^*)$$

is convex. Stationarity therefore implies that $\mathbf{x}^*$ globally minimizes this Lagrangian:

$$\mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) \geq \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) \quad \forall \mathbf{x}.$$

Now take any feasible $\mathbf{x}$. Since $\lambda_i^* \geq 0$ and $g_i(\mathbf{x}) \leq 0$, while $a_j(\mathbf{x}) = 0$,

$$\mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) \leq h(\mathbf{x}).$$

At $\mathbf{x}^*$, complementary slackness and equality feasibility give

$$\mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = h(\mathbf{x}^*).$$

Combining the three inequalities,

$$h(\mathbf{x}) \geq \mathcal{L}(\mathbf{x}, \lambda^*, \nu^*) \geq \mathcal{L}(\mathbf{x}^*, \lambda^*, \nu^*) = h(\mathbf{x}^*).$$

Hence $\mathbf{x}^*$ is globally optimal. □

Complementary slackness has a useful interpretation:

$$\lambda_i^* > 0 \implies g_i(\mathbf{x}^*) = 0.$$

Thus a constraint can carry a positive multiplier only if it is tight at the optimum.

Example 2.28 (A one-dimensional KKT calculation). Consider

$$\min_x x^2 \quad \text{subject to} \quad x \geq 1.$$

Write the constraint as

$$g(x) = 1 - x \leq 0.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda) = x^2 + \lambda(1 - x), \quad \lambda \geq 0.$$

The KKT conditions are

$$x \geq 1, \quad \lambda \geq 0, \quad 2x - \lambda = 0, \quad \lambda(1 - x) = 0.$$

If $\lambda = 0$, stationarity gives $x = 0$, which is infeasible. Hence the constraint must be active:

$$x^* = 1.$$

Stationarity then gives

$$\lambda^* = 2.$$

Thus the constrained optimum is $x^* = 1$, even though the unconstrained minimizer of x^2 is 0.

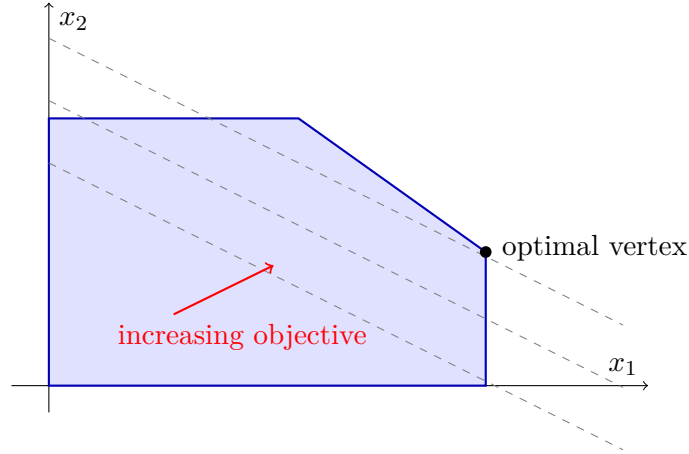


Figure 2.5: An LP optimizes a linear objective over a polyhedral feasible region. When an optimum exists on a bounded polytope, at least one optimal solution is a vertex.

2.4 Linear Programming

A linear program (LP) optimizes a linear objective over a feasible set defined by linear constraints. LPs are among the most important optimization problems in algorithms, economics, operations research, and game theory.

One convenient primal form is

$$\begin{aligned}
 \text{(P)} \quad & \max_{\mathbf{x} \in \mathbb{R}^n} \quad \mathbf{c}^\top \mathbf{x} \\
 & \text{subject to} \quad A\mathbf{x} \leq \mathbf{b}, \\
 & \quad \mathbf{x} \geq \mathbf{0},
 \end{aligned} \tag{2.2}$$

where $A \in \mathbb{R}^{m \times n}$, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{c} \in \mathbb{R}^n$.

The feasible region of an LP is a polyhedron. When bounded, it is a polytope.

Many equivalent LP forms are used in the literature. Equalities can be represented by two inequalities, unrestricted variables can be written as differences of nonnegative variables, and minimization can be converted to maximization by negating the objective.

2.4.1 The dual linear program

Associated with (2.2) is the dual problem

$$\begin{aligned}
 \text{(D)} \quad & \min_{\mathbf{y} \in \mathbb{R}^m} \quad \mathbf{b}^\top \mathbf{y} \\
 & \text{subject to} \quad A^\top \mathbf{y} \geq \mathbf{c}, \\
 & \quad \mathbf{y} \geq \mathbf{0}.
 \end{aligned} \tag{2.3}$$

The primal has n variables and m inequality constraints; the dual has m variables and n inequality constraints. The dual variables can be interpreted as prices or shadow values attached to the primal constraints.

2.4.2 Weak duality

Theorem 2.29 (Weak duality). *For every primal-feasible $\mathbf{x}$ and dual-feasible $\mathbf{y}$,*

$$\boxed{\mathbf{c}^\top \mathbf{x} \leq \mathbf{b}^\top \mathbf{y}.}$$

Proof. Dual feasibility gives

$$A^\top \mathbf{y} \geq \mathbf{c}.$$

Since $\mathbf{x} \geq 0$, multiplying componentwise and summing yields

$$\mathbf{x}^\top A^\top \mathbf{y} \geq \mathbf{x}^\top \mathbf{c} = \mathbf{c}^\top \mathbf{x}.$$

Primal feasibility gives $A\mathbf{x} \leq \mathbf{b}$. Since $\mathbf{y} \geq 0$,

$$\mathbf{y}^\top A\mathbf{x} \leq \mathbf{y}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{y}.$$

Finally,

$$\mathbf{x}^\top A^\top \mathbf{y} = \mathbf{y}^\top A\mathbf{x}.$$

Therefore

$$\mathbf{c}^\top \mathbf{x} \leq \mathbf{x}^\top A^\top \mathbf{y} = \mathbf{y}^\top A\mathbf{x} \leq \mathbf{b}^\top \mathbf{y}.$$

□

Weak duality already has an important algorithmic interpretation: every feasible dual solution provides an upper bound on the value of every feasible primal solution.

2.4.3 Strong duality

The remarkable fact for LPs is that, under the usual feasibility/boundedness assumptions, the best such upper bound is exact.

Theorem 2.30 (Strong duality for linear programming). *If the primal LP has a finite optimal value, then the dual has an optimal solution and*

$$\max_{\mathbf{x} \text{ feasible}} \mathbf{c}^\top \mathbf{x} = \min_{\mathbf{y} \text{ feasible}} \mathbf{b}^\top \mathbf{y}.$$

Equivalently, whenever both sides admit optimal solutions $\mathbf{x}^*, \mathbf{y}^*$,

$$\mathbf{c}^\top \mathbf{x}^* = \mathbf{b}^\top \mathbf{y}^*.$$

We will use strong duality as a standard theorem. There are several proofs, for example through the separating-hyperplane theorem, Farkas' lemma, or the simplex method.

2.4.4 Complementary slackness

Strong duality gives an especially useful way to recognize optimal primal–dual pairs.

Theorem 2.31 (Complementary slackness). *Let $\mathbf{x}^*$ be primal feasible and $\mathbf{y}^*$ dual feasible. They are both optimal if and only if*

$$y_i^*(b_i - (A\mathbf{x}^*)_i) = 0 \quad i = 1, \dots, m,$$

and

$$x_j^*((A^\top \mathbf{y}^*)_j - c_j) = 0 \quad j = 1, \dots, n.$$

Proof. For any primal/dual feasible pair,

$$\begin{aligned} \mathbf{b}^\top \mathbf{y} - \mathbf{c}^\top \mathbf{x} &= \mathbf{y}^\top (\mathbf{b} - A\mathbf{x}) + \mathbf{x}^\top (A^\top \mathbf{y} - \mathbf{c}) \\ &= \sum_{i=1}^m y_i (b_i - (A\mathbf{x})_i) + \sum_{j=1}^n x_j ((A^\top \mathbf{y})_j - c_j). \end{aligned}$$

Every term on the right-hand side is nonnegative by primal and dual feasibility.

If $\mathbf{x}^*, \mathbf{y}^*$ are optimal, strong duality says the left-hand side is zero. A sum of nonnegative terms can equal zero only if every term equals zero, giving the two complementary-slackness conditions.

Conversely, if a feasible pair satisfies complementary slackness, then the displayed duality gap is zero:

$$\mathbf{b}^\top \mathbf{y}^* = \mathbf{c}^\top \mathbf{x}^*.$$

Weak duality then implies that neither solution can be improved, so both are optimal. $\square$

Complementary slackness says, informally:

- a dual variable can be positive only when the corresponding primal constraint is tight;
- a primal variable can be positive only when the corresponding dual constraint is tight.

Example 2.32 (Primal, dual, and complementary slackness). Consider

$$\begin{aligned} \max_{x_1, x_2 \geq 0} \quad & 3x_1 + 2x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 4, \\ & x_1 \leq 2, \\ & x_2 \leq 3. \end{aligned}$$

The dual is

$$\begin{aligned} \min_{y_1, y_2, y_3 \geq 0} \quad & 4y_1 + 2y_2 + 3y_3 \\ \text{subject to} \quad & y_1 + y_2 \geq 3, \\ & y_1 + y_3 \geq 2. \end{aligned}$$

Take

$$\mathbf{x}^* = (2, 2), \quad \mathbf{y}^* = (2, 1, 0).$$

The primal value is

$$3(2) + 2(2) = 10,$$

and the dual value is

$$4(2) + 2(1) + 3(0) = 10.$$

Thus weak duality already certifies that both solutions are optimal.

Complementary slackness is also visible:

- the first and second primal constraints are tight, and their dual variables y_1^*, y_2^* may be positive;
- the third primal constraint is slack ($x_2^* = 2 < 3$), and indeed $y_3^* = 0$;
- both primal variables are positive, and both dual constraints hold with equality.

2.4.5 Algorithms for linear programming

The simplex method, developed by Dantzig, moves between vertices of the feasible polytope while improving the objective. It is extremely successful in practice, although natural pivoting rules can require exponentially many steps in the worst case; the Klee–Minty examples are a classical illustration [KM72].

A central theoretical question was whether LPs could be solved in time polynomial in the bit-length of the input. Khachiyan answered this affirmatively in 1979 by applying the ellipsoid method to linear programming [Kha79]. This was the first polynomial-time algorithm for general

LPs. Karmarkar subsequently introduced a new polynomial-time method in 1984 [Kar84]; his work was a landmark in the development of modern interior-point methods. For a detailed treatment of primal–dual interior-point algorithms, see [Wri97].

Thus, from a complexity perspective,

linear programming is solvable in polynomial time.

This fact¹ will be important later because many equilibrium, allocation, and mechanism-design problems can either be written directly as LPs or analyzed through their duals.

Remark 2.33. Polynomial time here refers to the standard bit-complexity model: the running time is polynomial in the dimensions of the LP and in the number of bits needed to encode its rational input. This should not be confused with the stronger notion of a *strongly polynomial* algorithm.

2.4.6 Exercises

Exercise 2.34. Consider the optimization problem

$$\max_{x_1, x_2, v} v$$

subject to

$$\begin{aligned} v &\leq 2x_1, & v &\leq x_2, \\ x_1 + x_2 &= 1, & x_1, x_2 &\geq 0. \end{aligned}$$

- (a) Solve the linear program and find the optimal values of x_1, x_2 , and v .
- (b) Explain why, at the optimum, the two constraints

$$v \leq 2x_1 \quad \text{and} \quad v \leq x_2$$

must both be tight.

Hint. If one of the two quantities $2x_1$ or x_2 is strictly larger than the other, can you change x_1, x_2 slightly to increase the smaller one?

Exercise 2.35. Consider the regularized optimization problem

$$\max_{\mathbf{x} \in \Delta_2} \left\{ \langle \mathbf{g}, \mathbf{x} \rangle - \frac{1}{2\eta} \|\mathbf{x}\|_2^2 \right\},$$

where

$$\mathbf{g} = (g_1, g_2) \in \mathbb{R}^2, \quad \eta > 0,$$

and

$$\Delta_2 = \left\{ \mathbf{x} \in \mathbb{R}^2 : x_1, x_2 \geq 0, x_1 + x_2 = 1 \right\}.$$

- (a) Write the Lagrangian for this optimization problem.

¹For this course, the above statement can be taken to be accurate. In general, saying that linear programming can be solved in polynomial time means that the running time grows efficiently with the size of the LP and with the size of the numbers used to describe it. There is also a stronger notion, called *strongly polynomial time*, where the running time depends only on the number of variables and constraints, and not on the numerical size of the coefficients.

(b) Assuming that the optimal solution satisfies

$$x_1^* > 0, \quad x_2^* > 0,$$

write down the first-order optimality conditions.

(c) Use these conditions to derive a relation between $x_1^* - x_2^*$ and $g_1 - g_2$.

Hint. Use a Lagrange multiplier for the constraint

$$x_1 + x_2 = 1.$$

Under the interior assumption, the nonnegativity constraints are inactive.

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